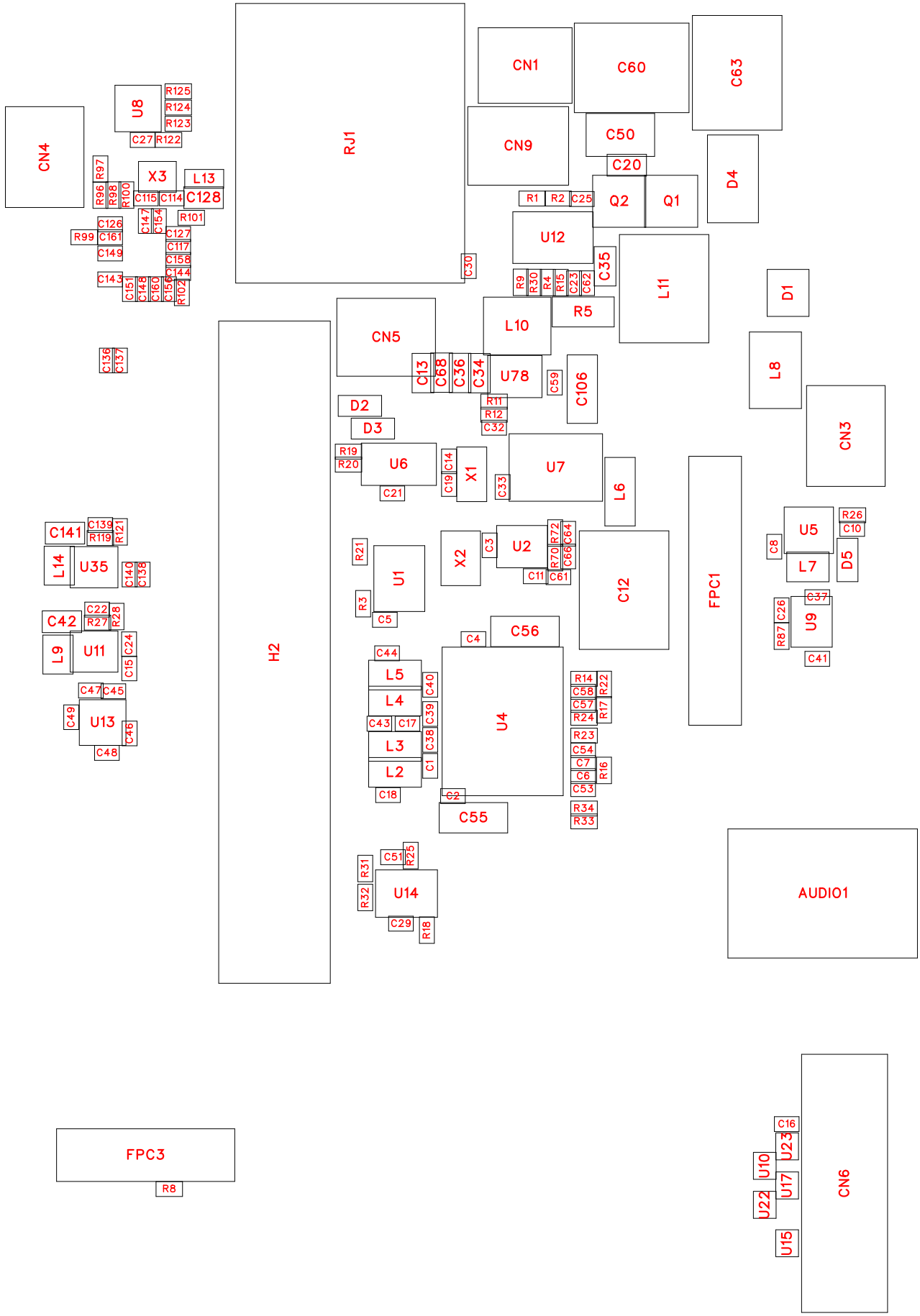
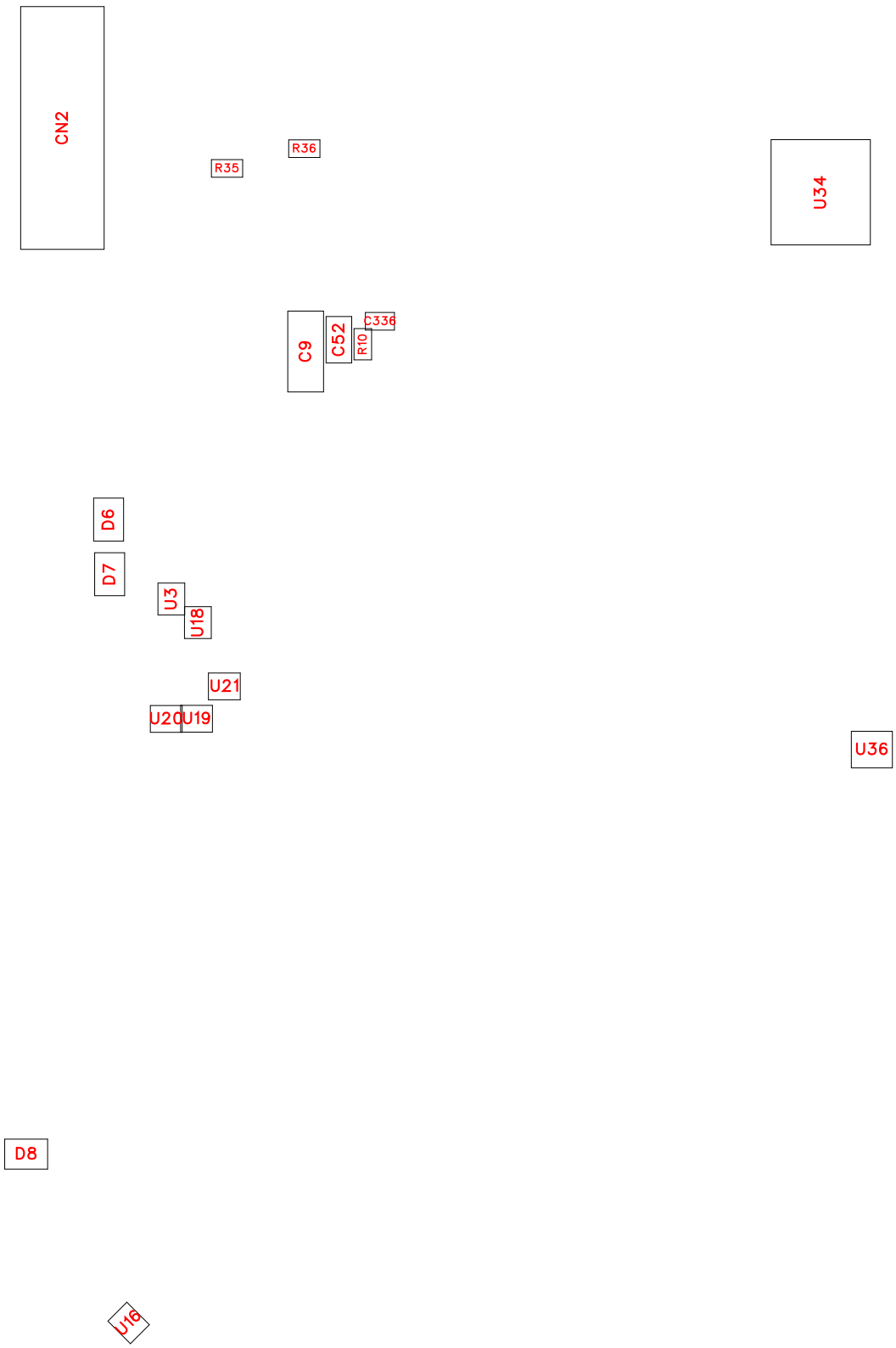

CN4
U8
R125
R124
R123
C27
R122
X3
L13
C128
R97
R96
R98
R100
C115
C114
C126
C161
C149
C147
C154
R101
C127
C117
C158
C144
C143
C151
C148
C160
C156
R102
C136
C137
RJ1
CN1
C60
C63
CN9
C50
C20
D4
R1
R2
C25
Q2
Q1
U12
C30
C35
L11
D1
R9
R30
R4
R15
C23
C62
R5
CN5
L10
L8
C13
C68
C36
C34
U78
C59
C106
CN3
R11
R12
C32
D2
D3
R19
R20
U6
C21
C14
C19
X1
C33
U7
L6
C141
C139
R119
R121
L14
U35
C140
C138
U5
R26
C10
C8
L7
D5
C37
U9
R87
C26
C41
C42
C22
R27
R28
L9
U11
C24
C15
C47
C45
C49
U13
C46
C48
H2
R21
U1
R3
C5
X2
C3
U2
R72
C64
R70
C66
C11
C61
C12
FPC1
C4
C56
C44
L5
L4
L3
L2
C40
C39
C38
C1
C43
C17
C18
U4
C2
C55
R14
C58
R22
C57
R17
R24
R23
C54
C7
R16
C6
C53
R34
R33
AUDIO1
C51
R25
R31
R32
U14
C29
R18
FPC3
R8
C16
U23
U10
U17
U22
U15
CN6


CN2
R35
R36
U34
C9
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R10
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D6
D7
U3
U18
U21
U20
U19
U36
D8
U16

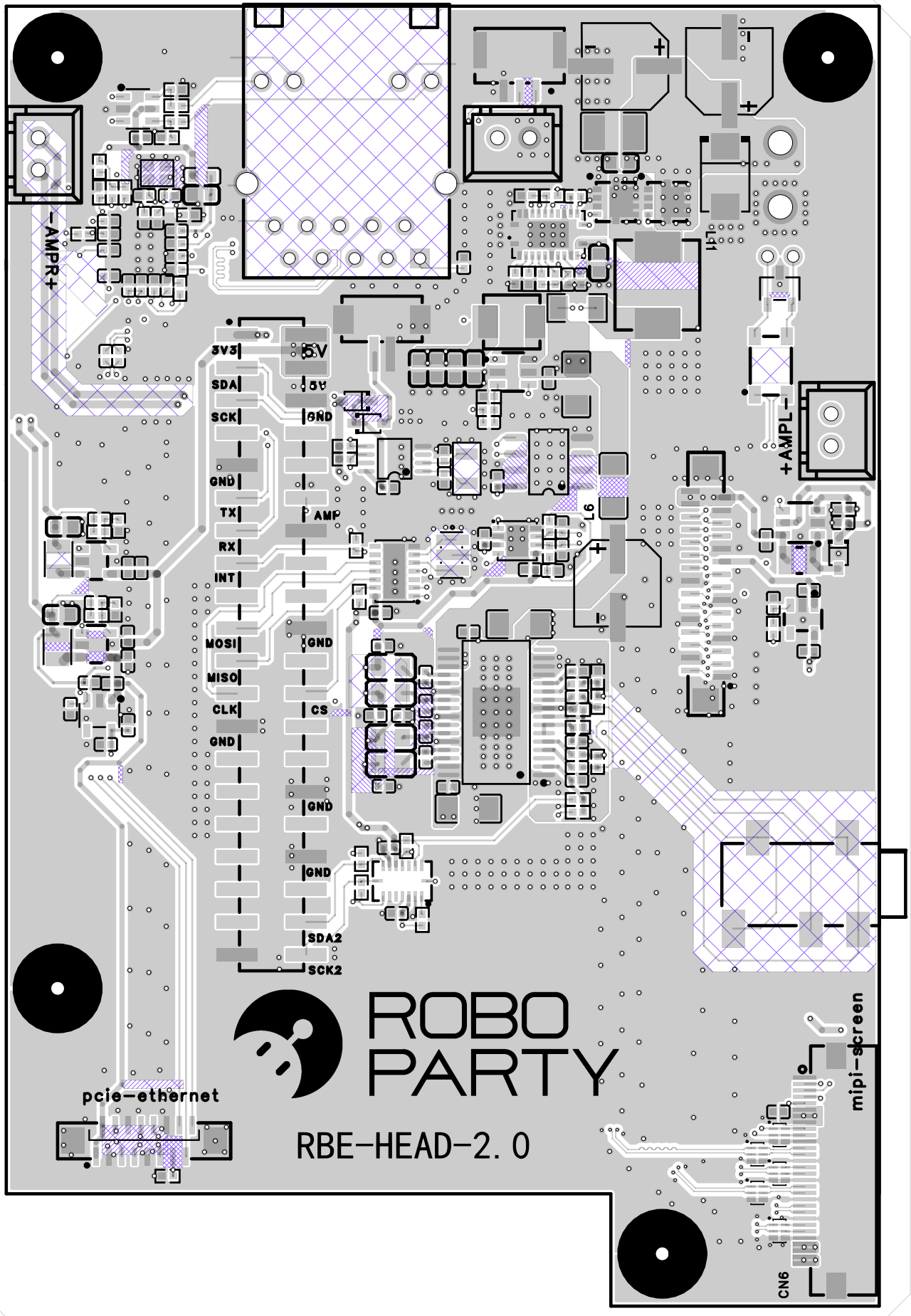
No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
1	1	PJ-327E-5A	AUDIO1	AUDIO-SMD_PJ-327E-5A		PJ-327E-5A	Hanyoung, קוריאה	C145812	LCSC
2	6	220nF	C1,C2,C4,C38,C39,C40	C0402	220nF	GRM155R61H224KE01D	muRata(Murata)	C5159775	LCSC
3	26	100nF	C3,C5,C11,C16,C21,C23,C24,C25,C27,C29,C30,C33,C47,C48,C51,C59,C136,C137,C140,C148,C149,C151,C158,C160,C161,C336	C0402	100nF	CC0402KRX7R9BB104,CL05B104KB54PNC,FCC0402B104K160AT	FOJAN(FOJAN) Samsung S AGE YAGEO)	C131394,C307331,C5137468	LCSC
4	4	3.3uF	C6,C7,C57,C58	C0402	3.3uF	C1005X5R1A335KTJ00E	TDK	C695126	LCSC
5	4	10uF	C8,C15,C126,C138	C0402	10uF	CL05A106MP5NUNC,GRM155R61A106ME11D	muRata(Murata MS Samsung)	C315248,C408132	LCSC
6	4	47uF	C9,C55,C56,C106	C1206	47uF	HGC1206R5476M160NSPJ	inocera(הואל קמקטע שם קרמיקה [מקטע])	C7472975	LCSC
7	12	1uF	C10,C37,C41,C45,C46,C54,C117,C143,C144,C147,C154,C156	C0402	1uF	CL05A105KA5NQNC	SUNG(Samsung)	C52923	LCSC
8	1	220uF	C12	CAP-SMD_BD6.3-L6.6-W6.6-FD-1	220uF	MA16V220M6X8	RR(IERR מקטע שם[רני [מקטע שם])	C46550461	LCSC
9	3	100nF	C13,C20,C35	C0603	100nF	CC0603KRX7R0BB104	YAGEO(YAGEO)	C113803	LCSC
10	4	12pF	C14,C19,C114,C115	C0402	12pF	0402CG120J500NT,CC0402JRNPO9BN120	FH(FH A GE YAGEO)	C1547,C106201	LCSC
11	7	1nF	C17,C18,C26,C43,C44,C49,C53	C0402	1nF	CC0402KRX7R9BB102,GRM1555C1H102JA01D	muRata(Murata AGE נו [מקטע שם] רני [מקטע שם])	C76947,C106205	LCSC
12	3	22pF	C22,C32,C139	C0402	22pF	CC0402JRNPO9BN220	YAGEO(YAGEO)	C106203	LCSC
13	6	22uF	C34,C36,C42,C52,C128,C141	C0603	22uF	GRM188R61A226ME15D,HGC0603R5226M160NTHJ	inocera(הואל קרמיקה [מקטע] Murata uRat)	C84419,C22367828	LCSC
14	1	4.7uF	C50	C1210	4.7uF	FS32X475K101EGG	A(מקטעPSA [קרמיקה [מקטע])	C381466	LCSC
15	2	22uF	C60,C63	CAP-SMD_BD6.3-L6.6-W6.6-FD-A	22uF	2201010609TR	NJCON(NJCON)	C19270543	LCSC
16	1	47nF	C61	C0402	47nF	CC0402KRX7R9BB473	YAGEO(YAGEO)	C272875	LCSC

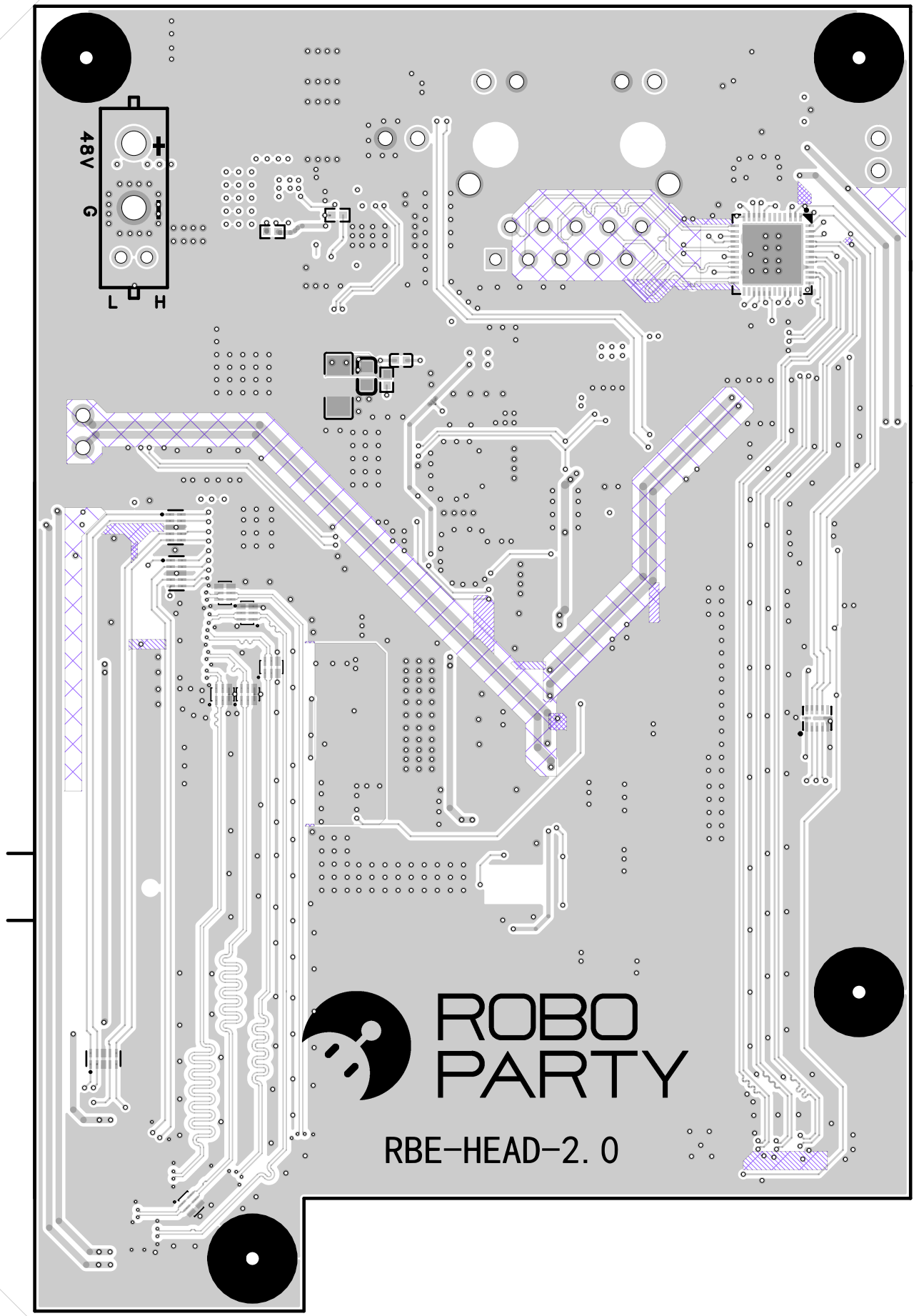
17	2	2.2uF	C62,C127	C0402	2.2uF	CC0402KRX5R6BB225,CL05A225KO5NQNC	SUNG(Samsung AGE YAGEO)	C170151,C326606	LCSC
18	2	47pF	C64,C66	C0402	47pF	CC0402JRNPO9BN470	YAGEO(YAGEO)	C60137	LCSC
19	1	10uF	C68	C0603	10uF	FCC0603X106K100CT	FOJAN(FOJAN)	C5137565	LCSC
20	1	WAFER-GH1.25-3PWB	CN1	CONN-TH_3P-P1.25_GH1.25		WAFER-GH1.25-3PWB	XUNPU(XUNPU)	C3029378	LCSC
21	1	XT30(2+2)PB-M.G.B	CN2	CONN-TH_XT30PB-M		XT30(2+2)PB-M.G.B	ASS(AMASS [מקטע שם] מה [מקטע שם])	C19268029	LCSC
22	3	XH-2A	CN3,CN4,CN9	CONN-TH_XH-2A		XH-2A	E(BOOMELE)	C20079	LCSC
23	1	A1252WV-S-2P	CN5	CONN-SMD_A1252WV-S-2P		A1252WV-S-2P	[מקטע שם] CJT מחבר [מקטע]	C400040	LCSC
24	1	KH-FG0.5-H2.0-30PIN	CN6	FPC-SMD_KH-FG0.5-H2.0-30PIN		KH-FG0.5-H2.0-30PIN	elm(Kinghelm [מקטע שם] שוב, [מקטע])	C2797216	LCSC
25	1	PESD1CAN	D1	SOT-23-3_L2.9-W1.6-P1.90-LS2.8-BR		PESD1CAN	BORN מוליך למחצה	C343993	LCSC
26	2	1N5819WS	D2,D3	SOD-323_L1.8-W1.3-LS2.5-RD		1N5819WS	ttech(Hottech [מקטע שם] טאי [מקטע שם])	C191023	LCSC
27	1	SMBJ60A	D4	SMB_L4.5-W3.7-LS5.4-RD		SMBJ60A	R+O(R+O)	C49066953	LCSC
28	1	PMEG4010BEA,115	D5	SOD-323_L1.8-W1.3-LS2.5-RD		PMEG4010BEA,115	xperia([מקטע שם] אין [מקטע שם])	C36975	LCSC
29	3	RCLAMP0524P	D6,D7,D8	DFN2510-10_L2.5-W1.0-P0.50-BL		RCLAMP0524P	T PUBLIC(TECH PUBLIC)	C907837	LCSC
30	1	FPC-0.5AZF2-30PLCR5	FPC1	FPC-SMD_FPC-0.5AZF2-30PLCR5		FPC-0.5AZF2-30PLCR5	XUNPU(XUNPU)	C54796568	LCSC
31	1	FPC-0.5AZF2-16PLCR5	FPC3	FPC-SMD_FPC-0.5AZF2-16PLCR5		FPC-0.5AZF2-16PLCR5	XUNPU(XUNPU)	C54796561	LCSC
32	1	HB-PH9-254220PB2GOW	H2	CONN-SMD_HB-PH9-254220PB2GOW		HB-PH9-254220PB2GOW	HANBO(HANBO)	C42396895	LCSC
33	4	MPZ2012S101ATD25	L2,L3,L4,L5	L0805		MPZ2012S101ATD25	TDK	C307776	LCSC
34	1	BLM31PG121SN1L	L6	L1206		BLM31PG121SN1L	muRata(Murata)	C85841	LCSC
35	1	10uH	L7	IND-SMD_L2.5-W2.0	10uH	FTC252012S100MBCA	ng(מקטע Cjiang [שם] חשמל [מקטע])	C5832376	LCSC

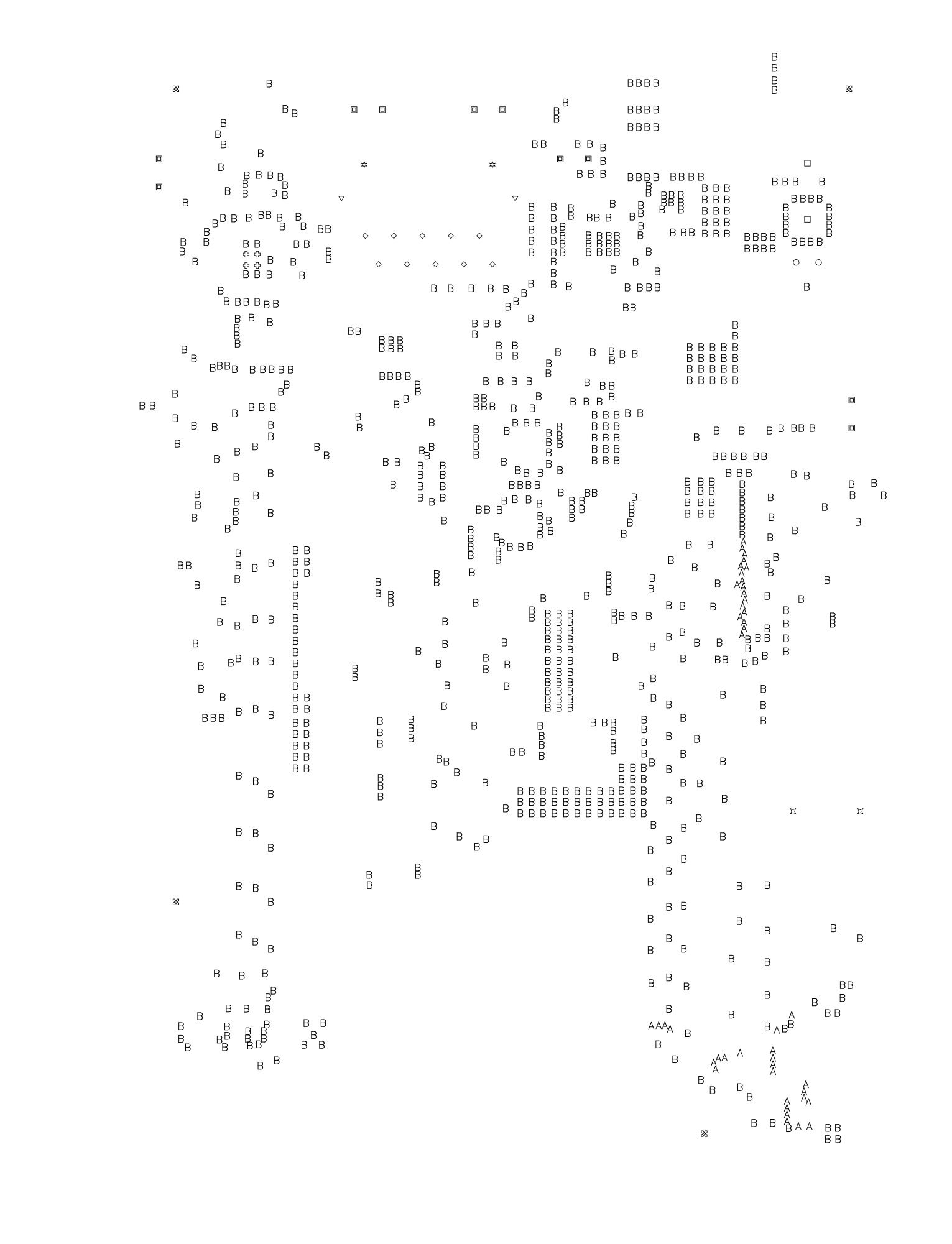
36	1	ACT45B-101-2P-TL003	L8	FILTER-SMD_4P-L4.5-W3.2-TL_ACT45B		ACT45B-101-2P-TL003	TDK	C88056	LCSC
37	2	2.2uH	L9,L14	IND-SMD_L2.5-W2.0_WPN252012H	2.2uH	WPN252012H2R2MT	Sunlord(Sunlord)	C98350	LCSC
38	1	3.3uH	L10	IND-SMD_L4.4-W4.2_SPM4020T	3.3uH	PCR0420-3R3-M	RR(IIERR[מקטע שם]רי[מקטע שם])	C53896857	LCSC
39	1	10uH	L11	IND-SMD_L7.3-W6.8	10uH	FXL0630-100-M	ng(מקטעCjiang [שם]חשמל[מקטע])	C167223	LCSC
40	1	BLM18PG121SN1D	L13	L0603		BLM18PG121SN1D	muRata(Murata)	C14709	LCSC
41	2	SQSA84CENW-T1_GE3	Q1,Q2	WDFN-5_L3.1-W3.1-P0.65-BL_1		SQSA84CENW-T1_GE3	VISHAY(Vishay)	C20346183	LCSC
42	1	6.49kΩ	R1	R0402	6.49kΩ	RT0402BRD076K49L	YAGEO(YAGEO)	C852898	LCSC
43	1	200kΩ	R2	R0402	200kΩ	0402WGF2003TCE	ROYAL(UNI-ROYAL)	C25764	LCSC
44	12	4.7kΩ	R3,R19,R20,R21,R31,R32,R98,R99,R100,R102,R122,R125	R0402	4.7kΩ	RC0402FR-074K7L	YAGEO(YAGEO)	C105871	LCSC
45	6	10kΩ	R4,R8,R14,R18,R25,R34	R0402	10kΩ	0402WGF1002TCE,FRC0402F1002TS,RC0402FR-0710KL	FOJAN(FOJAN)-R(UNI-ROYAL AGE YAGEO)	C25744,C60490,C2906861	LCSC
46	1	5mΩ	R5	R1206	5mΩ	RE1206F1R005	RR(IIERR[מקטע שם]רי[מקטע שם])	C46961735	LCSC
47	1	41.2kΩ	R9	R0402	41.2kΩ	FRC0402F4122TS	FOJAN(FOJAN)	C2998053	LCSC
48	2	100kΩ	R10,R23	R0402	100kΩ	AC0402FR-07100KL,THC02G1003BT	FH(FH AGE YAGEO)	C144809,C47120018	LCSC
49	2	15kΩ	R11,R96	R0402	15kΩ	FRC0402F1502TS,RC0402FR-0715KL	FOJAN(FOJAN)AGE[מקטע שם]רי[מקטע שם])	C114761,C2909322	LCSC
50	1	110kΩ	R12	R0402	110kΩ	FRC0402F1103TS	FOJAN(FOJAN)	C2909311	LCSC
51	1	48.7kΩ	R15	R0402	48.7kΩ	RT0402BRD0748K7L	YAGEO(YAGEO)	C852805	LCSC
52	6	1kΩ	R16,R17,R22,R33,R87,R97	R0402	1kΩ	RC0402FR-071KL	YAGEO(YAGEO)	C106235	LCSC
53	1	20kΩ	R24	R0402	20kΩ	FRC0402F2002TS	FOJAN(FOJAN)	C2906867	LCSC
54	1	5.1Ω	R26	R0402	5.1Ω	FRC0402F5R10TS	FOJAN(FOJAN)	C2909363	LCSC
55	2	30kΩ	R27,R119	R0402	30kΩ	RC0402FR-0730KL	YAGEO(YAGEO)	C138008	LCSC

56	1	6.8kΩ	R28	R0402	6.8kΩ	FRC0402F6801TS	FOJAN(FOJAN)	C2906876	LCSC
57	1	61.9kΩ	R30	R0402	61.9kΩ	FRC0402F6192TS	FOJAN(FOJAN)	C2933106	LCSC
58	2	0Ω	R35,R36	R0402	0Ω	FRC0402F0000TS	FOJAN(FOJAN)	C2906858	LCSC
59	2	60.4Ω	R70,R72	R0402	60.4Ω	RC0402FR-0760R4L	YAGEO(YAGEO)	C137954	LCSC
60	1	2.49kΩ	R101	R0402	2.49kΩ	RC0402FR-072K49L	YAGEO(YAGEO)	C145071	LCSC
61	1	49.9kΩ	R121	R0402	49.9kΩ	RC0402FR-0749K9L	YAGEO(YAGEO)	C137969	LCSC
62	2	470Ω	R123,R124	R0402	470Ω	0402WGF4700TCE	UNI-ROYAL(UNI-ROYAL)	C25117	LCSC
63	1	HR911130A	RJ1	RJ45-TH_HR911130A		HR911130A	NRUN(NRUN)	C54408	LCSC
64	1	MCP2518FDT-E/QBB	U1	VDFN-14_L4.5-W3.0-P0.65-BL-EP		MCP2518FDT-E/QBB	HIP(Microchip)	C621395	LCSC
65	1	TCAN1044VDRBRQ1	U2	VSON-8_L3.0-W3.0-P0.65-BR-EP		TCAN1044VDRBRQ1	TI(Texas Instruments)	C5215851	LCSC
66	11	RCLAMP0522P	U3,U10,U15,U16,U17,U18,U19,U20,U21,U22,U23	DFN-6_L1.6-W1.0-P0.50-BL		RCLAMP0522P	T PUBLIC(TECH PUBLIC)	C2937020	LCSC
67	1	TPA3118D2DAPR	U4	HTSSOP-32_L11.0-W6.1-P0.65-LS8.1-BL-EP		TPA3118D2DAPR	TI(Texas Instruments)	C46497	LCSC
68	1	MT9201	U5	SOT-23-6_L2.9-W1.6-P0.95-LS2.8-BR		MT9201	Xi'an Aerospace Minxin	C182966	LCSC
69	1	PCF8563TS/5,118	U6	TSSOP-8_L3.0-W3.0-P0.65-LS4.9-BL		PCF8563TS/5,118	NXP(NXP)	C27397	LCSC
70	1	ATECC608A-SSHDA-T	U7	SOIC-8_L4.9-W3.9-P1.27-LS6.0-BL		ATECC608A-SSHDA-T	HIP(Microchip)	C616024	LCSC
71	1	SN74LVC2G17DBVR	U8	SOT-23-6_L2.9-W1.6-P0.95-LS2.8-BL		SN74LVC2G17DBVR	TI(Texas Instruments)	C10429	LCSC
72	1	TPS7A2028PDBVR	U9	SOT-23-5_L2.9-W1.6-P0.95-LS2.8-BR		TPS7A2028PDBVR	TI(Texas Instruments)	C2869847	LCSC
73	2	ETA3409S2F	U11,U35	SOT-23-5_L2.9-W1.6-P0.95-LS2.8-BL		ETA3409S2F	ETA(ETA)	C7465553	LCSC
74	1	LM5148RGYR	U12	VQFN-24_L5.5-W3.5-P0.50-BL-EP-2		LM5148RGYR	TI(Texas Instruments)	C7470701	LCSC

75	1	TPS7A2018PDBVR	U13	TSOT-23-5_L2.9-W1.6-P0.95-LS2.8-BL		TPS7A2018PDBVR	TI(Texas Instruments)	C963430	LCSC
76	1	BMI088	U14	LGA-16_L4.5-W3.0-P0.50-BL		BMI088	Bosch(Bosch)	C194919	LCSC
77	1	RTL8125BG-CG	U34	TQFN-48_L6.0-W6.0-P0.40-BL-EP4.5		RTL8125BG-CG	ALTEK(ALTEK)	C3013605	LCSC
78	1	RS0104YUTQH12	U36	QFN-12_L2.0-W1.7-P0.40-BL		RS0104YUTQH12	RUNIC(RUNIC)	C5372422	LCSC
79	1	SY8113IADC	U78	TSOT-23-6_L2.9-W1.6-P0.95-LS2.8-BL		SY8113IADC	ergy(Silergy)	C479075	LCSC
80	1	32.768kHz	X1	OSC-SMD_L3.2-W1.5	32.768kHz	SC-32S32.768kHz20PPM12.5pF	Seiko(Seiko)	C97606	LCSC
81	1	20MHz	X2	CRYSTAL-SMD_3P-L3.2-W1.3-P1.2-BL	20MHz	CSTNE20M0V530000R0	muRata(Murata)	C882604	LCSC
82	1	25MHz	X3	CRYSTAL-SMD_4P-L2.0-W1.6-BL-A	25MHz	XC21M4-25.000-F12NLDTL	HCI(HCI)	C3034312	LCSC


-AMPR+
+AMPL-
3V3
SDA
SCK
GND
TX
RX
INT
MOSI
MISO
CLK
GND
5V
5V
GND
AMP
GND
CS
GND
GND
SDA2
SCK2
L11
L6
ROBO
PARTY
RBE-HEAD-2.0
pcie-ethernet
mipi-screen
CN6


48V
G
L
H
ROBO
PARTY
RBE-HEAD-2.0



ROBOSKIN | RP1-derived engineering record

English file: 06_PCB_PDF_HEAD_2026-09-16.pdf
Status: derivative identification; NOT a manufacturing release.

Existing dimensions, tolerances, material and revision remain authoritative.
Source engineering graphics and original attribution are retained.
Technical notes may remain in the source language.
Only searchable project part IDs on original pages are replaced.
Outlined/raster labels require the original CAD authoring application.

Original drawing pages: 10
Searchable ID replacements: 0
Source: Roboparty/rp1_hardware
Hardware licence: CERN-OHL-W-2.0 where applicable; see LICENSES.md.
Original path and SHA256: electronic_skin/branding/file_manifest.json

Part aliases in this document:

ROBOSKIN - מקרא עברי לטקסט השרטוט

המידות, החומרים והסבולות שבשרטוט נשמרו. המספר משמאל מציין עמוד ושדה.

p3 T1529	Hanyoung, קוריאה
p3 T1032	Murata
p3 T0805	FOJAN
p3 T0440	Samsung
p3 T0715	YAGEO
p3 T0644	הואה [מקטע שם]
p3 T1166	קרמיקה [מקטע]
p3 T0968	JIERR [מקטע שם]
p3 T1165	רוי [מקטע שם]
p3 T1534	FH
p3 T0713	גו [מקטע שם]
p3 T0876	ג׳ו [מקטע שם]
p3 T0542	PSA [מקטע שם]
p3 T1512	קרמיקה [מקטע]
p3 T1098	NJCON
p4 T0440	Samsung
p4 T0715	YAGEO
p4 T0805	FOJAN
p4 T1364	XUNPU
p4 T1331	AMASS [מקטע שם]
p4 T0993	סה [מקטע שם]
p4 T0660	BOOMELE
p4 T1498	CJT [מקטע שם]
p4 T0973	מחבר [מקטע]

ROBOSKINמקרא עברי לטקסט השרטוט -

המידות, החומרים והסבולות שבשרטוט נשמרו. המספר משמאל מציין עמוד ושדה.

p4 T1441	זהב / ג׳ין [מקטע]
p4 T1328	Kinghelmמקטע שם] [
p4 T0524	BORN
p4 T0643	מוליך למחצה
p4 T0692	Hottechמקטע שם] [
p4 T1107	טאי [מקטע שם]
p4 T0793	R+O
p4 T0790	אן [מקטע שם]
p4 T0465	שי [מקטע שם]
p4 T0687	TECH PUBLIC
p4 T1100	HANBO
p4 T1032	Murata
p4 T1497	Cjiangמקטע שם] [
p4 T1172	חשמל [מקטע]
p5 T1531	Sunlord
p5 T0968	JIERRמקטע שם] [
p5 T1165	רוי [מקטע שם]
p5 T1497	Cjiangמקטע שם] [
p5 T1172	חשמל [מקטע]
p5 T1032	Murata
p5 T0787	Vishay
p5 T0715	YAGEO
p5 T0665	UNI-ROYAL
p5 T0805	FOJAN

ROBOSKIN - מקרא עברי לטקסט השרטוט

המידות, החומרים והסבולות שבשרטוט נשמרו. המספר משמאל מציין עמוד ושדה.

p5 T1534	FH
p5 T0713	גו [מקטע שם]
p5 T0876	ג׳ו [מקטע שם]
p6 T0805	FOJAN
p6 T0715	YAGEO
p6 T0665	UNI-ROYAL
p6 T1099	האן [מקטע שם]
p6 T0517	רן [מקטע שם]
p6 T1277	Microchip
p6 T0904	Texas Instruments
p6 T0687	TECH PUBLIC
p6 T1355	Xi’an Aerospace Minxin
p6 T0916	NXP
p6 T1446	ETA
p7 T0904	Texas Instruments
p7 T0659	Bosch
p7 T1165	רוי [מקטע שם]
p7 T1016	יו [מקטע שם]
p7 T1109	RUNIC
p7 T1217	[מקטע שם Silergy]
p7 T1034	גייה [מקטע שם]
p7 T1245	Seiko
p7 T1032	Murata
p7 T1033	HCI